Surprise-gated learning in a spiking network: one-page summary

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The question. When we hear a rare tone among frequent ones, neurons fire more. Is that just tired neurons (adaptation), or is the brain learning what to expect and signalling surprise (prediction error)? I also wanted to know what is the *smallest* piece of machinery that turns a network that only adapts into one that learns expectations quickly.

What I built. An open, fully reproducible spiking neural network (120 neurons, Brian2) in which every mechanism can be switched on and off on its own:

- two kinds of adaptation;
- a predictive-coding core with learned predictions and error neurons;
- my proposal, a **surprise gate**. After a prediction error it briefly speeds up learning, raises sensory gain and opens several candidate guesses about what comes next.

I wrote down every prediction and pass rule before running each experiment (pre-registration), ran 20 random seeds per condition, and published every result, including the predictions that failed. The repository has 79 unit tests, one command per figure, and cached, resumable runs.

What I found.

1. **Adaptation is not enough.** Adaptation alone reproduces the familiar "bigger response to the rare tone", but all of it is repetition suppression. The prediction-error component is zero, and there is no response at all to a *missing* tone. This holds even when adaptation is made 10 times slower.
2. **Surprise gating buys speed, not new abilities.** Two tests separate adaptation from expectation: an unexpected repeat (ABAB**AA**) and a missing tone. Adaptation gets the repeat backwards (−2.8 spikes). The surprise gate gets it right after normal training (+7.6 [4.0, 11.2]). The plain predictive core also gets there, but only with about 3 times more training. Against a perfect Bayesian learner on the same task, the gate learns in about the same number of examples (5 vs 6); the plain core needs about 8 times more. Ablations show that passing both tests needs prediction-error neurons, learning and the gate together.
3. **Honest limits.** One-shot learning, a predicted drop-off ordering and "accepting randomness" were *not* supported, and a first attempt at fixing the noise problem (a global precision signal) failed by silencing all error. When I slowed adaptation 3 times, the effects first disappeared; re-tuning by my original rules (a 15% change to three weights) brought them back. At 10 times slower, only a weak repeat effect returns.

Why it matters for AI. Standard training treats each example in much the same way, whether or not it was surprising. This model is a small worked example of a different rule: **learn fast exactly when the world surprises you, and keep a few competing hypotheses open while you do**. That is the core of sample-efficient, continual learning: adapt to a change in a handful of examples without overwriting everything else. The failures are informative too. The gate alone never stops reacting to pure noise: in random input it keeps making confident guesses that a perfect learner would not. My first fix, a single global "how surprising has the world been" signal, silenced all errors and broke learning. So precision has to be specific to what is being predicted. That is the next thing I am building.

Paper (10 pages) and code: github.com/Aryan-sk6/SNNresearch, `paper/paper.pdf`.